PHY-112 | Principles of Physics-2

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Spring 2025 | Class #0

Department of Mathematics & Natural Sciences

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Introduction

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Introduction
Assessment Rubric

How will your performance be assessed?

- **05%** $\Longrightarrow$ Attendance
- **15%** $\Longrightarrow$ Averaging 4 Quizzes (15 marks each)
- **15%** $\Longrightarrow$ Averaging 4 Assignments (15 marks each)
- **20%** $\Longrightarrow$ Mid-term examination (20 marks)
- **10%** $\Longrightarrow$ Laboratory experiments (10 marks)
- **35%** $\Longrightarrow$ Final examination (35 marks)

NOTE: Course is to be central. There will be No Bonus Assessments! You have one shot at this. Make it count!

Syllabus & Prerequisites

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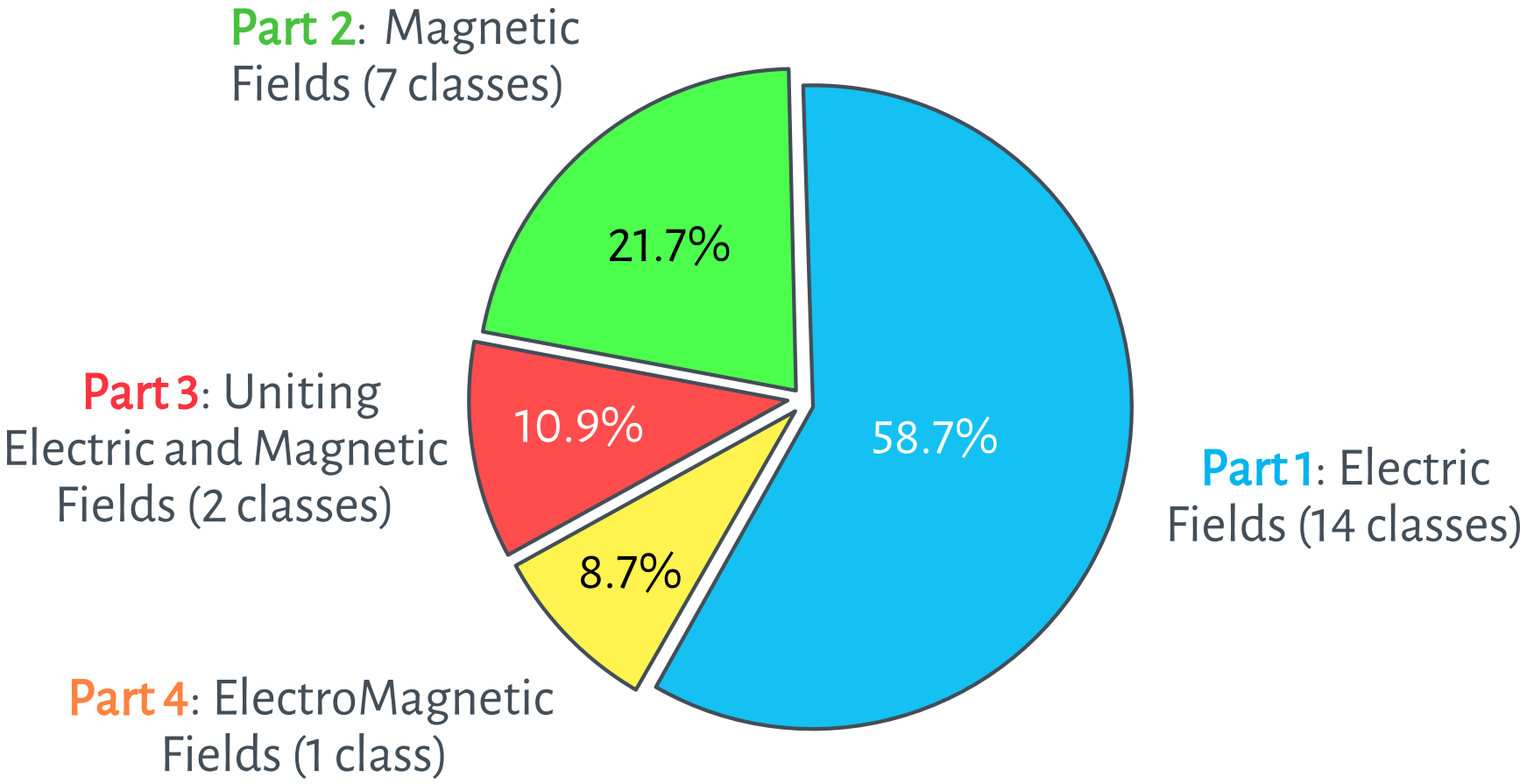
Syllabus

How the course is structured



Syllabus

How the course is structured

Part 1: Electric Fields (6 classes)

- Electric Charge: Micro and Macroscopic. Properties of Charges
- Electric Charge Distribution: Discrete and Continuous
- Electrostatic Forces
- Coulomb's Law of Electrostatics
- How is Electric Field Created?
- Introduction to $\vec{E}$-Field: Field Lines and Intensity
- Charged Particles in $\vec{E}$-Field: Straight and Parabolic Deflection
- Gauss's Law of Electrostatics
- Electric Flux: Uniform and Non-Uniform
- Applications of Gauss's Law: Line, Plane Charges
- List the 1st Maxwell's Equation

Syllabus

How the course is structured

Part 1: Electric Fields (5 classes)

- Work Done due to Electrostatic Force
- Electric Potential Energy
- Electric Potential Intuition
- Electric Potential Measurements for Charge Distributions: Point, Line, Plane. (Integration Heavy)
- Equipotential Surfaces
- Capacitance and Capacitors: With and without Dielectrics
- Energy Stored in $\vec{E}$-Field

Mid-Term will be up until this.

Syllabus
How the course is structured

Part 1: Electric Fields (3 classes)

- What's the point of an Electric Circuit?
- Electric Current: Microscopic View
- Ohm's Law and Conducting Materials
- Electromotive Force (EMF)
- Power in(out)put from Electric Circuits
- Electric Circuit Introduction and Elements
- Solving DC Circuits: Kirchoff's Laws
- Transient Series *RC* Circuits: Charging and Discharging of a Capacitor

Syllabus
How the course is structured

Part 2: Magnetic Fields (7 classes)

- How is Magnetic Field Created? Intuition with examples.
- Biot-Savart Law
- Magnetic Field Intensity Measurements for Current Distributions: Straight Wire, Wire Loop.
- Magnetic Field of a Current Loop and Electromagnet
- Magnetic Force Caused by $\vec{B}$-Fields on Wires: Single Point Charge, Straight Wire, Wire Loop
- Charged Particles in $\vec{B}$-Field: Circular Deflection, Cyclotron
- Gauss's Law for Magnetostatics: Magnetic Flux
- Ampère's Law and how to use it
- List the 2nd Maxwell's Equation

Syllabus
How the course is structured

Part 3: Uniting Electric & Magnetic Fields (2 or 3 classes)

- ElectroMagnetic Induction: Induced EMF and Induced Current
- Faraday's Law of Induction & Lenz's Law
- Inductance and Inductors
- List the 3rd Maxwell's Equation

Part 4: ElectroMagnetic Fields (1 of 0.5 class)

- Displacement Current and Ampère-Maxwell Law
- List the 4th Maxwell's Equation
- Generate and Propagate the ElectroMagnetic Wave: LIGHT

Final will contain 10-15% of Mid Syllabus.

Prerequisites

What You need to ace the course!

- Pre-requisite: **PHY-111**
  - Vector Resolution: Geometric and Analytical
    - Breaking any vector down
  - Vector Algebra: Addition and Subtraction
    - Superposition of forces and fields
  - Vector Multiplication: Dot and Cross-Product
    - Electric and Magnetic Flux (Dot)
    - Biot-Savart Law and Magnetic Forces (Cross)
  - Intuition of Fields: Vector and Scalar Fields in 3D
    - Electric and Magnetic Fields
  - Work Done and Conservation of Energy
    - Electric Potential Energy
    - Motion of charged particles

What to and not to expect!

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Theory or Math?

Spot the difference

Number in, Number out!

An electric field of strength 15000 N C^{-1} is inside a parallel-plate capacitor of 1.5 mm spacing. An electron is released from rest at the negative plate. What is the electron's speed when it reaches the positive plate?

Same thing but with no calculator!

An electric field of strength E is inside a parallel-plate capacitor of d spacing. A proton is released from rest at the negative plate. What is the proton's speed when it reaches the negative plate?

Same thing but with common sense!

An electric field of strength E is inside a parallel-plate capacitor of d spacing. Explain why when a proton or electron is released from the rest from the midpoint of the plate separation, the electron will hit one of the plates before the proton does.

Resources

What to use?

- In-Person Classes (Main Food)
- Consultation Hours (Supplementary Vitamins)
- Reference Books (Will be provided in Slack)
- Exercise Problem List (Will be provided in Slack)
- Student Tutor
- Each Other
- The Internet

Resources
How to use?

- Pay *effective* Attention to in-person classes every small detail matters. Do not leave out everything for consultation hour
- Test Yourself constantly: Solve practice problems
- Do *more* of what works
- Do *less* of what does not
- Do not let things pile up!!
- Respect your time, brain, and willpower
- Rinse and Repeat

That is it for today!